

ROBOTS EMPOWERING Law And Order



► Using various sensors, motors and output channels, robots can feed on information in real time and aid law enforcement officers in times of need

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Protectors of law dedicate their lives to keep our cities safe. However, it is not possible for them to reach every place in need, in time. In this regard, robots can bring great benefits. As these are becoming more interactive and intelligent, robots have become helping hands for police personnel.

Countries across the globe have been reinforcing their police forces with robots, and deployments have shown great progress. Looking at some of these successful deployment stories, Indian police forces can also consider investing in robotic assistants for their benefit.

How robots can strengthen law enforcement

Robots can be put to use in different applications on the streets. With the amalgamation of various sensors, motors and output channels, these can feed on necessary information in real time and provide assistance as required.

Main sensory components of these machines are lidars, auto-sonic sensors, touch sensors, cameras with computer vision, microphones, input touchscreens and so on. Inputs picked up by the sensors are processed in a microcontroller (MCU) to create relative outputs.

Most robots today are being used for street-level assistance of citizens, like filing reports and complaints,

performing security scans, generating tickets, assisting with directions and more. These also send real-time alerts and notifications to police control centres in case of incidents or emergencies, provide information regarding medical aid, gas stations, repair shops and so on.

Implementation of robots can bring many advantages, the foremost being safety and preparedness of the human police workforce. Many officers often get victimised in unsuspecting incidents. Robots, in these conditions, can alert police forces beforehand through their security scan capabilities, and help in preventing mishaps.

Robots can also deliver real-time traffic and parking information, helping in improving urban traffic conditions. Moreover, these are easily accessible to citizens for any kind of support, in turn, reducing wait times at police stations and load on officers.

There are many more benefits that robots can bring to administratives that can, in turn, reduce life risk and expenses, and improve overall citizen wellness.

Deployment stories

Robotics in Indian law enforcement is yet to see proper deployments. However, solution providers in the country are working towards developing machines that can benefit police departments in the future. For instance, Hyderabad-based H-bot Robotics has developed a policing robot that the company claims to be ready for deployment, and is tailored to Indian conditions.

H-bot Robotics' robots can facilitate interactive support to citizens, conduct road surveillance, help in traffic management and detect explosives. These will be available commercially soon, for